Huan Peng

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Education

KTH Royal Institute of Technology — **Stockholm, Sweden**
M.Sc. in Systems, Control and Robotics — *Aug 2024 – Jun 2026*

- GPA: 4.94/5.0
- Selected Coursework: Nonlinear Control, Machine Learning Theory, Hybrid and Embedded Control Systems, Applied Nonlinear Optimization, Model Predictive Control, Cyber-Physical Security in Time-Critical Systems.
- Thesis: Equilibrium Robustness for Non-Convex Minimax Problems with Uncertain Constraints.
  Examiner: Prof. Karl Henrik Johansson; Supervisor: Prof. Guanpu Chen.

Tianjin University — **Tianjin, China**
B.Eng. in Automation — *Sep 2020 – Jun 2024*

- GPA: 91.5/100
- Thesis: On the Vulnerability of Cyber-Physical Systems Under Hybrid Attacks.
  Supervisor: Prof. Zhiqiang Zuo.

Research Interest

Randomized and robust optimization, Distributed optimization, Network games, Statistical learning theory

Publications

[1] H. Peng, G. Chen, and K. H. Johansson, "A Scenario Approach to the Robustness of Nonconvex–Nonconcave Minimax Problems," accepted by *IFAC World Congress 2026* (Preprint: arXiv:2511.15606).

[2] H. Peng, G. Chen, G. Belgioioso, and K. H. Johansson, "Distributed Algorithm for Robust Wardrop Equilibrium in Uncertain Aggregative Congestion Games," submitted to *IEEE Transactions on Control of Network Systems* (Preprint: arXiv:2606.01594).

Research Experience

Research Assistant — **Aug 2024 – June 2026**
Under the supervision of Prof. Guanpu Chen and Prof. Karl Henrik Johansson — *DCS Division, EECS KTH, Sweden*

- Proposed a robust distributed algorithm to compute equilibria under structural uncertainties, proving its asymptotic convergence in the framework of nonlinear control theory. The effectiveness of the proposed method was demonstrated in charging control of plug-in electric vehicles.
- Established robustness guarantees for nonconvex–nonconcave minimax problems leveraging data-driven techniques, namely the "scenario approach". The validity of these guarantees was demonstrated in power system applications.
- Derived a novel theoretical bound on the average number of mistakes for scenario optimization to ensure risk-averse decision-making. This result provided key theoretical support for safety-critical applications such as MPC and finance.

Teaching Experience

Laboratory Teaching Assistant — **Aug 2025 – Dec 2025**
KTH EL1000, EL1010 & EL1020, Automatic Control, Basic Course — *Part-time*

- Supervised bachelor students in laboratory sessions on classical control topics, including PID control, pole placement, and lead–lag compensation.
- Guided students through system modeling, controller design, and real-time implementation.
- Evaluated student performance and determined lab pass/fail results based on technical understanding and execution.

Awards

- KTH One-Year Scholarship, KTH Royal Institute of Technology — Apr 2025
- Merit Student Scholarship, Tianjin University — Dec 2021, 2022 & 2023
- Second Prize, Tianjin Undergraduate Electronic Design Contest (Texas Instruments cup) — Dec 2022
- First Prize, Tianjin College Student Mathematics Competition — Jul 2021

Skills

Programming: Python, MATLAB, C/C++
Tools: ROS, SIMULINK, Git, Docker
Languages: Chinese (Native), English (Fluent), Swedish (Basic)

Last updated: June 8, 2026